

Muhammed Ali Onurlucan1

ali.onurlucan@metu.edu.tr | malionurlucan.me | linkedin.com/in/mali0

Ankara, Türkiye

EDUCATION

Middle East Technical University

BSc in Mathematics, CGPA: 3.17

2023–current

INTERNSHIP / WORK EXPERIENCE

Türkiye National Observatory – Research Intern

June 2026–July 2026

Investigated the mathematical foundations of Adaptive Optics (Zernike polynomial expansion and wavefront reconstruction) and Space Situational Awareness (SSA) models utilizing TLE/SGP4 orbit propagation under the supervision of Prof. Dr. Cahit Yeşilyaprak.

Galaxy Rocket Team – Team Member

2025–current

Developing and analyzing mathematical models related to flight dynamics and propulsion systems using MATLAB/Simulink.

PREPTTEST – Question Writing and Exam Preparation

2023–current

Preparing and editing exam questions for AP Calculus, SAT, Töc, and Bocconi on the Preptest platform.

ACADEMIC ACTIVITIES

Higher Structures Seminar, Kadri İlker Berktaş

October 2025

Attending weekly seminars on advanced topics in higher category theory.

Turkish Mathematical Society (TMD) – Member

2025–current

Active member of the national society promoting mathematical research, education, and collaboration in Turkey.

The Nesin Mathematics Village

August 2024

Participated in intensive mathematics lessons on Surreal Numbers and Axiom of Choice.

PROJECTS

CS50P: Introduction to Programming with Python

2023

Developed a password manager for secure password handling. Source code and documentation available on GitHub.

CERTIFICATIONS & COURSES

PHYS-209-ENx: Introduction to Astrophysics

August 2024

Explored celestial mechanics, galactic structures, and observational techniques.

ANU-ASTRO1x: Greatest Unsolved Mysteries of the Universe

December 2024

Studied stellar evolution, black holes, and cosmology.

SCHOLARSHIPS

TÜBİTAK 2205 – Undergraduate Scholarship Program

2022–current

Awarded a national scholarship recognizing academic excellence and potential in STEM fields.

SKILLS

Programming & Tools: Python, MATLAB/Simulink, LaTeX, Git/GitHub, Wolfram Mathematica

Languages: Turkish (Native), English (Fluent)